Numerical Simulation and Trajectory Analysis of Charged Particle Deflection in Gridded Magnetic Ion Thrusters

Ulykpan Bolekbai
github.com/ulykpanbolekbai

October 2026

Abstract

This paper presents an interactive two-dimensional framework for studying the motion of charged particles in the magnetic deflection zone of a gridded ion thruster. The model, written in Python with the Pygame library, injects electrons or singly charged Argon ions from an ionization chamber into a uniform magnetic field, where their motion is governed by the Lorentz force and advanced with a per-frame Euler scheme. The program shows the cyclotron radius, period and Lorentz force in real time while the user varies the field strength, injection angle and particle species. The displayed quantities are checked against closed-form expressions, and a geometric criterion for the return of the beam through the entrance boundary of the field zone is derived. An offline replication of the integrator shows that explicit Euler increases the particle speed by a factor $\sqrt{1+(\omega_c \Delta t)^2}$ per frame, which is negligible for heavy ions but severe for electrons at the default settings. A speed-conserving rotation (Boris) update with sub-stepping is proposed as a remedy. The tool is intended as an educational and preliminary-design aid for understanding beam steering in electric propulsion.

Keywords: ion thruster, Lorentz force, cyclotron motion, particle simulation, Euler method, Boris algorithm.

1 Introduction

Electric propulsion systems accelerate a propellant with electric and magnetic fields rather than chemical combustion. The exhaust velocity is much higher than for chemical rockets, so the specific impulse is high and the propellant mass needed for deep-space missions is small [1]. In a gridded ion thruster the propellant gas (commonly Xenon or Argon) is ionized in a discharge chamber, and the ions are extracted and accelerated through biased grids.

Magnetic fields play several roles in such devices: they confine the ionizing electrons, increasing ionization efficiency, and they can steer charged species. Knowing how a particle moves under a given magnetic flux density is therefore important for beam symmetry and for preventing erosion of walls and grids.

Alternative ion sources are also actively studied. Ionic-liquid ion sources with externally wetted emitters produce well-defined bipolar beams at modest voltages and have been proposed for micropropulsion [2]. Whatever the source, the extracted charged particles obey the same Lorentz-force dynamics once they enter a magnetic region, which motivates a simple and transparent simulation of that stage.

The contributions of this work are:

1. an interactive 2D simulator solving the motion of an electron or Argon ion in a uniform field in real time;

2. live diagnostics (cyclotron radius, period, force) and their verification against analytical results;

3. a geometric criterion for the return of the beam through the field-zone entrance;

4. a quantitative analysis of the error introduced by the explicit Euler integrator, and a proposed fix.

2 Physics and Mathematical Framework

2.1 Lorentz force dynamics

A particle of charge q and mass m moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ experiences

$$\mathbf{F} = q\,(\mathbf{v} \times \mathbf{B}). \tag{1}$$

The force is perpendicular to $\mathbf{v}$, so the field changes the direction of motion but does no work and the kinetic energy is constant. For $\mathbf{B} = B\,\hat{\mathbf{z}}$ pointing into the screen and motion in the xy plane, Eq. (1) gives

$$F_x = q\,v_y B, \qquad F_y = -q\,v_x B, \tag{2}$$

with magnitude $F = |q|vB\sin\alpha$, where α is the angle between $\mathbf{v}$ and $\mathbf{B}$. Because $\mathbf{B}$ is normal to the motion plane, $\alpha = 90^\circ$ always, and $F = |q|vB$. (The injection angle set by the slider is a different quantity and is denoted θ below.)

2.2 Cyclotron motion

In a uniform field the force acts as a centripetal force. Setting $|q|vB = mv^2/R$ gives

$$R = \frac{mv}{|q|B}, \tag{3}$$

$$\omega_c = \frac{|q|B}{m}, \tag{4}$$

$$T = \frac{2\pi m}{|q|B}. \tag{5}$$

The period does not depend on speed, while the radius is proportional to momentum. At equal speed and $|q|$, the radius scales with mass, so a heavy ion is deflected much less than an electron; the sense of rotation reverses with the sign of q.

2.3 Geometry of the field zone

In the simulator the field zone is a rectangle of width W and height H whose left edge coincides with the nozzle exit, so particles are born on its boundary. For injection normal to this edge ($\theta = 0$) the orbit is a circle of radius R centred on the edge line. Without any other force the particle completes a half circle and leaves through the same boundary, displaced by $2R$ from the injection point, provided the half circle stays inside the zone. With h the distance from the injection point to the top edge, this holds when $2R \le h$, i.e. for

$$B \ge B_{\mathrm{ret}} = \frac{2\,mv}{|q|\,h}. \tag{6}$$

For $B < B_{\mathrm{ret}}$ the particle leaves through the top of the zone with a deflection angle smaller than 180° and continues in a straight line. Thus the zone behaves as a reflector of the beam for strong fields and as a deflector for weak fields.

2.4 Numerical integration

Once per frame, with time step $\Delta t = 1$, the code updates velocity and then position:

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \frac{\mathbf{F}_n}{m}\,\Delta t, \tag{7}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v}_{n+1}\,\Delta t. \tag{8}$$

The force is evaluated only for particles inside the zone; outside, the particle moves in a straight line.

3 Implementation and Simulation Architecture

The program is a single Python module (`main.py`, about 400 lines) using Pygame, run at 60 FPS in an 800×600 window.

Geometry. The thruster is drawn with `pygame.draw` primitives: a propellant feed line, the ionization chamber with internal electrode lines, a converging nozzle, power lines, and the magnetic-field zone with a dot grid indicating $\mathbf{B}$ into the page (Fig. 1).

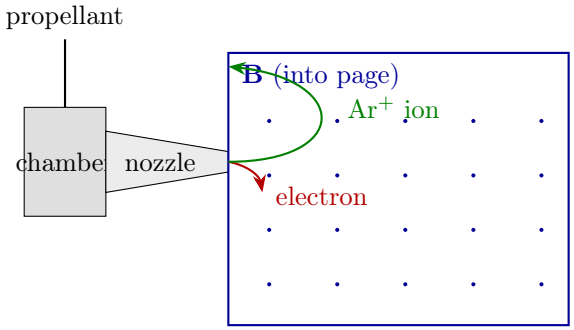


Figure 1: Schematic of the simulated layout (not to scale). Particles are injected normally into the field zone; for a strong field both species make a half turn and return through the entrance, the electron on a much smaller circle.

Particles. Two species are defined in scaled units (Table 1). A new particle is created every 100 ms (at most 80 live) at $x = x_0 = 220$ px, $y = 300 \pm 15$ px, i.e. on the left edge of the field zone, with speed $v_0 = 3.5$ px/frame directed at angle $\theta \in [-30°, 30°]$ to the x axis. A trail of the last 35 positions is drawn with fading brightness, and particles are removed when they leave the window.

The mass ratio of 20 is chosen for visibility; the real ratio $m_{\mathrm{Ar}}/m_e \approx 7.3 \times 10^4$ is far larger, so real electrons are deflected much more strongly relative to ions than in the simulator.

Code structure. Physics functions: `lorentz_force()`, `apply_magnetic_force()`, `calc_radius()`, `calc_period()`. Classes: `Slider`, `ToggleButton`, `Particle`, `ParticleSystem`. Rendering: `draw_thruster_body()`, `draw_field_zone()`, `draw_physics_panel()`, `draw_ui_panel()`.

Table 1: Main parameters of the simulation (scaled units).

Parameter	Value	Note
Electron mass, charge	0.05, -1	mass in simulation units ("u")
Argon ion mass, charge	1.0, $+1$	mass ratio 20:1
Speed v_0	3.5	px/frame
Time step Δt	1	one frame
B range (slider)	0.01–0.30	default 0.08
Injection angle θ	$-30°..+30°$	default 0°
Field zone	340×440 px	top-left at (220, 80)
Spawn interval / max particles	100 ms / 80	

4 Results

4.1 Verification of the displayed quantities

The HUD computes R, T and F from the constants m, q, v_0 and the current slider value B. For the default electron case ($m = 0.05$, $|q| = 1$, $v_0 = 3.5$, $B = 0.08$):

$$R = \frac{0.05 \times 3.5}{1 \times 0.08} = 2.19 \text{ px}, \quad T = \frac{2\pi \times 0.05}{1 \times 0.08} = 3.93 \text{ frames}, \quad F = 1 \times 3.5 \times 0.08 = 0.280,$$

which agree with the values on the screen (Table 2). For Argon at the same B: $R = 43.75$ px and $T = 78.5$ frames. Note that these panel values are the *ideal* ones for constant speed v_0; they do not use the actual instantaneous particle speed, and therefore they validate the formulas, not the integrator (see Section 5).

Table 2: Analytical results versus simulator HUD (electron, default parameters).

Quantity	Analytical	HUD	Unit
Radius R	2.19	2.19	px
Period T	3.93	3.93	frames
Force F	0.2800	0.2800	sim. units

4.2 Dependence on field strength and species

Figure 2 plots $R(B)$ for both species over the slider range, with the field-zone width $W = 340$ px for reference. The Argon radius equals 20 times the electron radius at every B. At $B = 0.01$ the Argon radius (350 px) is comparable to the zone width, whereas the electron radius never exceeds about 17.5 px, so electrons are always strongly bent.

With $y_0 = 300$ and $h = y_0 - 80 = 220$ px, Eq. (6) gives $B_{\text{ret}} = 0.032$ for Argon, so ions return to the entrance for $B \gtrsim 0.032$ and leave through the top for weaker fields. For the electron $B_{\text{ret}} = 1.6 \times 10^{-3}$ lies below the slider range, so (in the ideal motion) electrons always return.

5 Numerical Considerations

5.1 Explicit Euler for magnetic rotation

The exact solution in a static magnetic field is a rotation at constant speed. In complex notation $w = v_x + iv_y$ the exact step multiplies w by $e^{\mp i\omega_c \Delta t}$, while the explicit Euler update of Eq. (7)

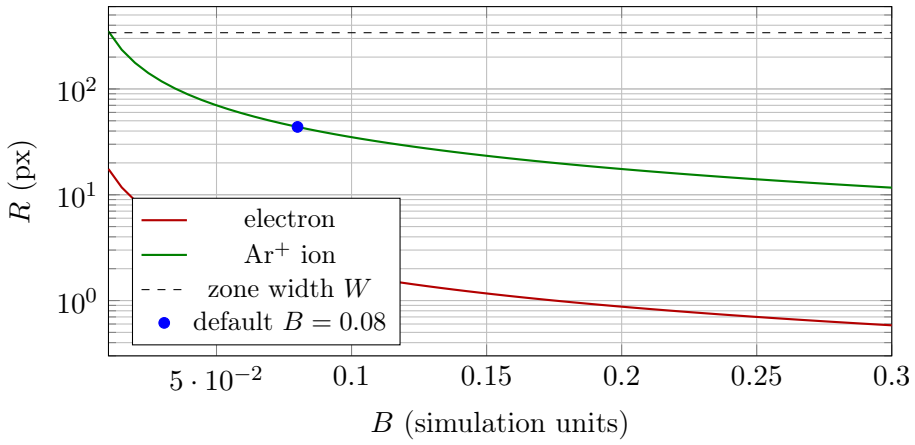


Figure 2: Cyclotron radius versus magnetic field for the two species at $v_0 = 3.5$.

multiplies it by $1 \mp i\omega_c \Delta t$, whose modulus is

$$|1 \mp i\omega_c \Delta t| = \sqrt{1 + (\omega_c \Delta t)^2} > 1. \tag{9}$$

The speed and kinetic energy therefore increase by this factor on every frame spent in the field, and the error is small only if $\omega_c \Delta t \ll 1$, i.e. $\Delta t \ll T/2\pi$. With $\omega_c = |q|B/m$ and $\Delta t = 1$:

- Argon, $B = 0.08$: $\omega_c \Delta t = 0.08$, factor 1.0032 per frame;
- electron, $B = 0.08$: $\omega_c \Delta t = 1.6$, factor 1.887 per frame.

In addition, the electron's orbital diameter $2R = 4.4$ px is comparable to its step length $v_0 \Delta t = 3.5$ px, so the circular orbit is not resolved at all (about 4 steps per period).

5.2 Offline replication

The update rules of `Particle.update()` were replicated in a separate script (spawn at $y_0 = 300$, $\theta = 0$, no random offset) and compared with a reference that rotates the velocity by the exact angle $-\omega_c \Delta t$ each frame (speed conserved). Table 3 summarizes the outcome.

Table 3: Explicit Euler (as in `main.py`) versus a speed-conserving rotation reference.

Case	Integrator	Frames in field	Speed after exit	Exit point
Ar$^+$, $B = 0.08$	Euler	38	3.95	left edge, $y \approx 182$
	reference	39	3.50	left edge, $y \approx 208$
Ar$^+$, $B = 0.04$	Euler	77	3.72	left edge, $y \approx 105$
	reference	78	3.50	left edge, $y \approx 120$
Ar$^+$, $B = 0.02$	Euler	91	3.56	top edge, $x \approx 366$
	reference	92	3.50	top edge, $x \approx 362$
electron, $B = 0.08$	Euler	2	12.46	left edge
	reference	1	3.50	left edge

For Argon the Euler error grows with the time spent in the field: the speed rises by $1.0032^{38} \approx 13\%$ at $B = 0.08$ (energy by about 28%) and the orbit spirals outward, but the qualitative behaviour (return through the entrance for $B \gtrsim 0.032$, exit through the top for weaker fields) is correct. For the electron the effect is dominant: the speed after the first two frames is 6.6 and 12.5 (factors 1.887 each), so the particle leaves the zone after two frames with 3.6 times its initial speed, i.e. about 12.7 times its initial kinetic energy, and then flies in a straight line. The

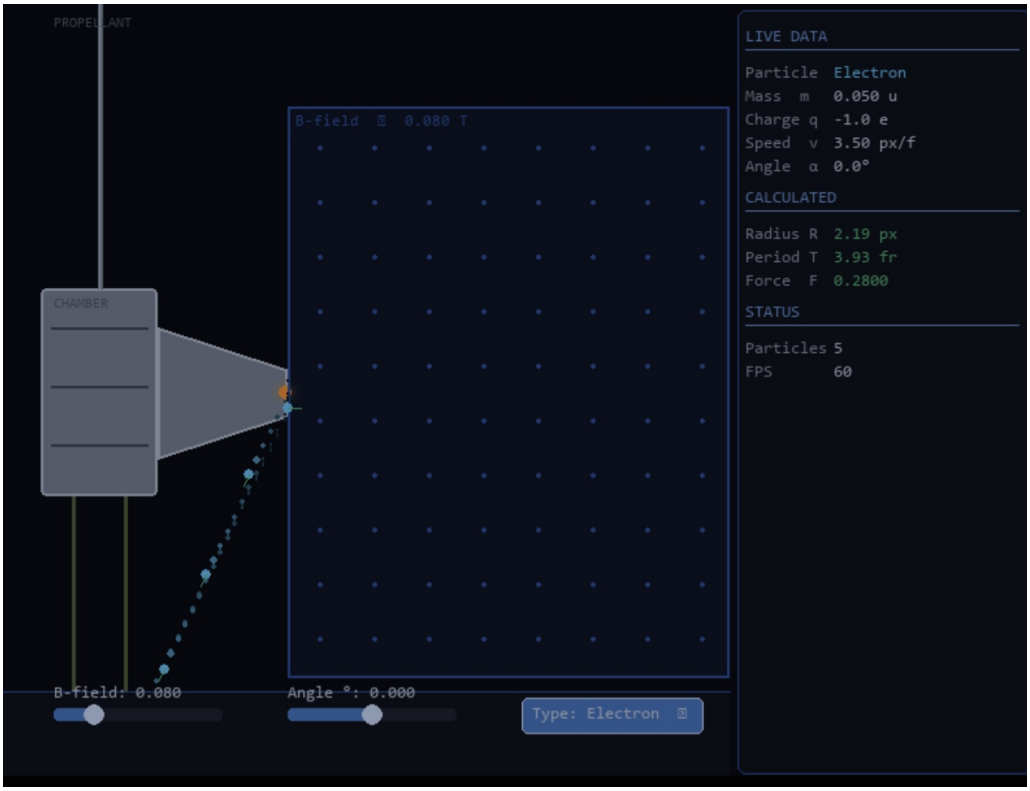


Figure 3: Screenshot of the simulator: nozzle, magnetic field zone, fading trails and the live-data panel.

nearly straight, long electron trails seen in the simulator are a numerical artefact of this energy gain and not the expected small circles of radius 2.19 px.

5.3 Proposed fix

Since a static magnetic field conserves speed, the velocity update can be replaced by an exact rotation of the velocity vector,

$$\begin{pmatrix} v_x \\ v_y \end{pmatrix}_{n+1} = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix}_{n}, \qquad \varphi = -\frac{qB}{m}\Delta t, \tag{10}$$

which for a pure magnetic field is equivalent to the Boris algorithm [3] and preserves $|\mathbf{v}|$ to rounding error. In Python:

```
def apply_magnetic_force(vx, vy, charge, b_field, mass, dt):
    phi = -charge * b_field / mass * dt
    c, s = math.cos(phi), math.sin(phi)
    return vx * c - vy * s, vx * s + vy * c
```

To also resolve the geometry of the electron orbit, each frame should be divided into N sub-steps with $\Delta t/N \lesssim T/20$; for the default electron ($T = 3.93$ frames) this means $N \approx 5$–10.

6 Discussion and Limitations

The model isolates the magnetic-deflection stage. It is two-dimensional, assumes a uniform field with sharp edges, and neglects space-charge interaction between particles, collisions, relativistic effects and the electrostatic acceleration grids that actually produce thrust in a gridded

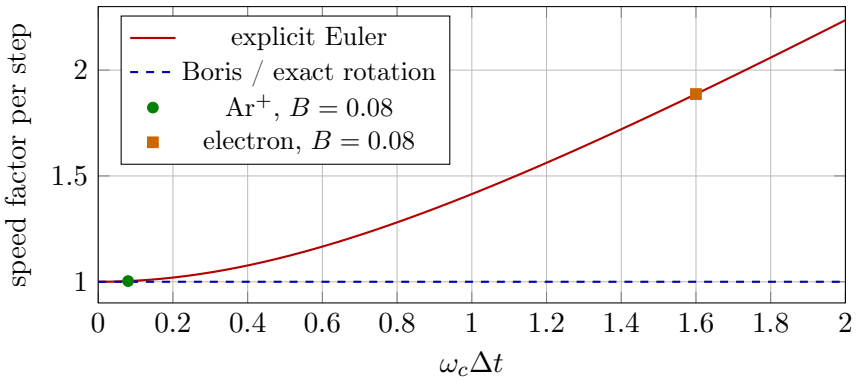


Figure 4: Per-step growth of speed in a pure magnetic field for explicit Euler versus an exact rotation.

thruster [1]. Units are scaled rather than SI and the mass ratio is artificially small. Moreover the field zone starts exactly at the nozzle exit, which makes the return-to-entrance behaviour of Eq. (6) appear for moderate fields; in a real thruster the field geometry would be designed differently. Within these limits the simulator reproduces the correct qualitative physics: $R = mv/(|q|B)$, opposite deflection for opposite charges, and much weaker deflection of heavy ions.

7 Conclusion and Future Work

An interactive 2D simulator of charged-particle deflection in a magnetic ion thruster was developed and checked against the analytical cyclotron relations. A geometric criterion for the beam returning through the field entrance was derived, and the explicit Euler integrator was shown to be adequate for heavy ions but to pump energy into light, strongly magnetized particles such as electrons. Future work includes:

1. adopting the rotation (Boris) update with sub-stepping as the default integrator;
2. adding an electrostatic acceleration grid with potential difference V, so that the exit speed follows $v = \sqrt{2|q|V/m}$;
3. including space-charge repulsion between particles;
4. moving from scaled units to SI values for Xenon and Argon;
5. using HUD values based on the actual instantaneous speed of the tracked particles;
6. extending the model to 3D and to non-uniform fields.

Code Availability

The source code is available at `https://github.com/ulykpanbolekbai`.

References

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